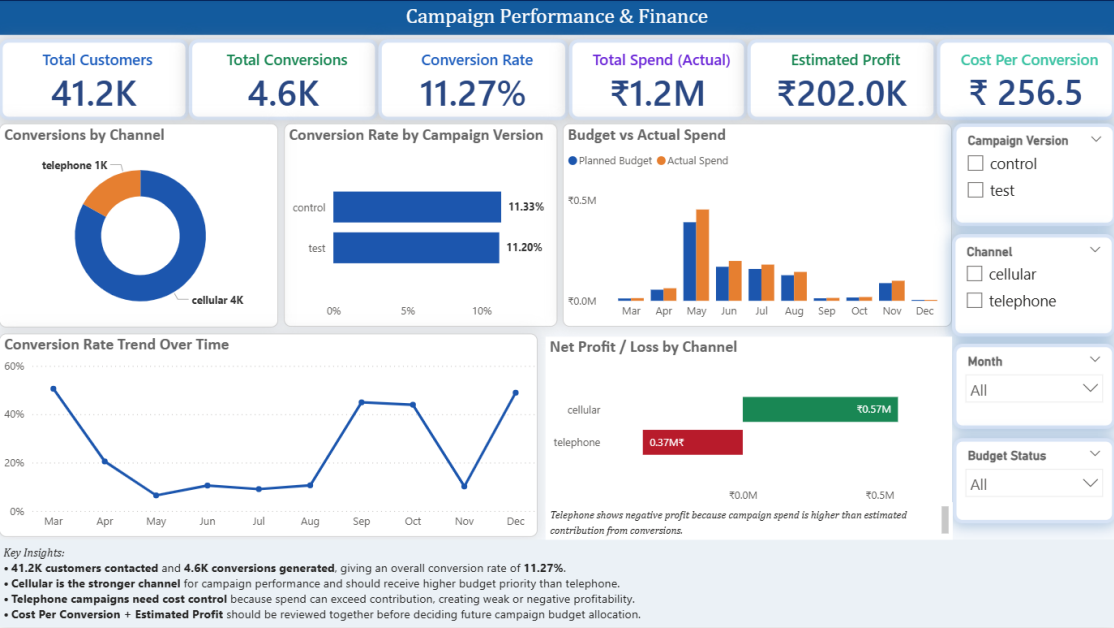


BFSI Campaign Analytics Project

AI-Assisted Bank Term Deposit Campaign Optimization

Python • SQL Server • Power BI • A/B Testing • Lead Scoring

- End-to-end analytics workflow built from raw banking data to executive dashboards.
- Focus: conversion, channel profitability, campaign testing, and lead prioritization.
- Created for junior data analyst portfolio and interview storytelling.



Banking campaign challenge

- The bank wants to improve term deposit subscriptions from phone-based marketing campaigns.
- Management needs to know which channel, campaign version, and customer segments perform best.
- Finance teams need visibility into actual spend, cost per conversion, and estimated profitability.
- Sales/CRM teams need a prioritized customer list for follow-up.

Business questions answered

Campaign

Control vs Test

Channel

**Cellular vs
Phone**

Finance

Spend & Profit

Customer

Lead Scoring

Decision

**Scale / Retest /
Reduce**

Inputs used in the project

01 UCI Bank Data

Real public bank marketing data used for customers, contacts and subscription outcome.

02 Campaign Cost File

Excel assumption table for planned cost, actual cost and contribution per conversion.

03 A/B Assignment

Synthetic control/test layer generated on the same customer records to demonstrate experiment workflow.

04 ML Scores

Conversion probabilities created in Python to prioritize customers.

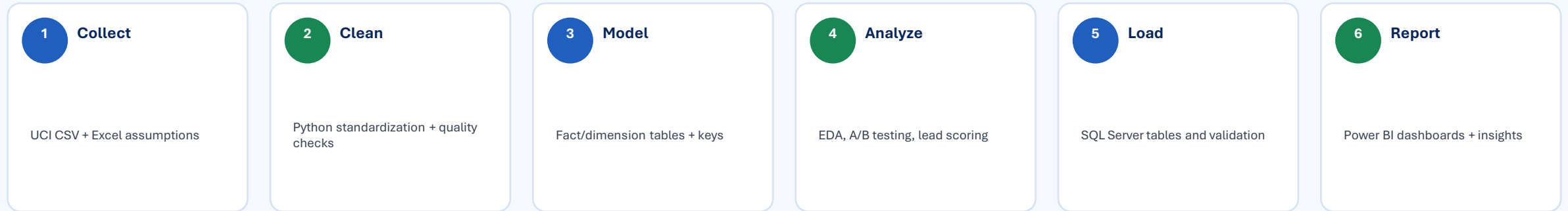
05 SQL Server

Clean CSV outputs loaded into a local SQL Server database for BI modeling.

06 Power BI

Final dashboard connected to model tables and DAX measures.

Transparency note: the original public dataset does not include campaign cost, true A/B variants, or original customer IDs. Those fields were created as documented assumptions for portfolio demonstration.



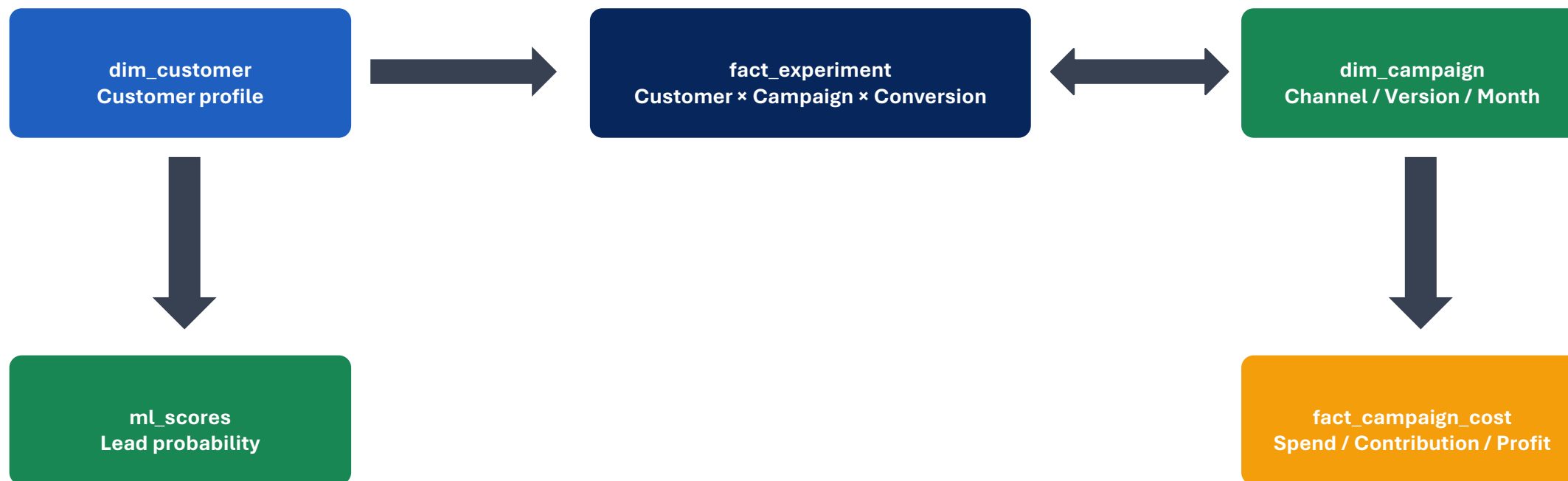
Python transformation outputs

- clean_bank_data.csv
- dim_customer.csv, dim_campaign.csv
- fact_experiment.csv, fact_campaign_cost.csv
- ab_test_result.csv, ml_scores.csv

Analytical questions answered

- Which campaign version performs better?
- Which channel is profitable?
- Which customers should be targeted first?
- Where should budget be controlled?

BI-ready relationship design



SQL Server was used as the analytical layer before Power BI. The model uses valid keys such as customer_id and campaign_id, instead of forcing unrelated tables together.

Data Quality

Removed duplicates, standardized text, created unknown-value flags.

A/B Testing

Compared control and test conversion rates using a two-proportion z-test.

Finance Analysis

Tracked actual spend, estimated contribution, profit, and cost per conversion.

Lead Scoring

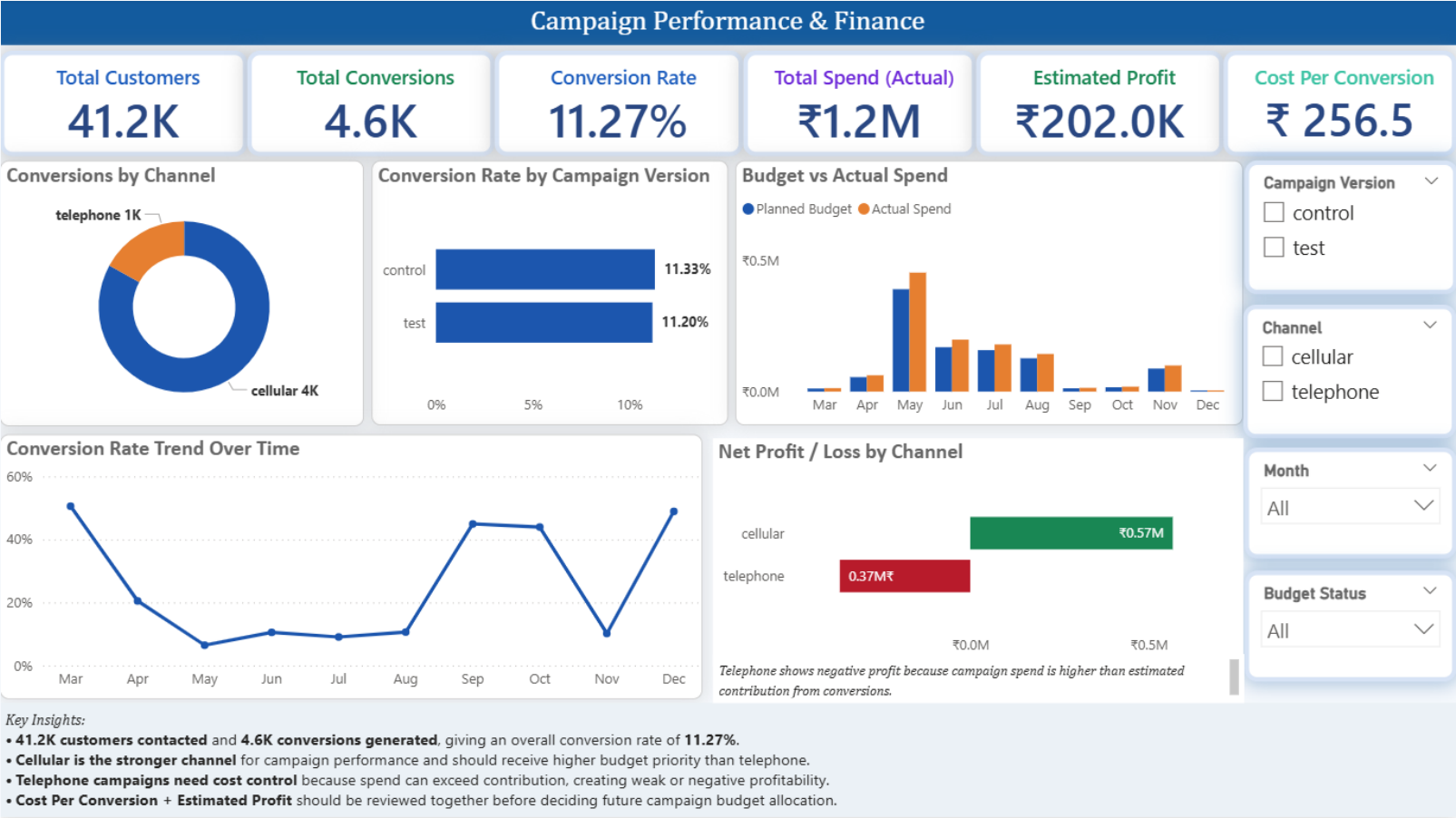
Used logistic regression to assign customer-level conversion probability.

Dashboarding

Built two executive pages with KPIs, slicers, charts and key insights.

AI Assistance

Used AI lightly for framing, debugging and concise insight wording.





Customers

41.2K

Conversions

4.6K

Conversion Rate

11.27%

Actual Spend

₹1.2M

Profit

₹202K

CPC

₹256

What management should do

- Prioritize cellular campaigns where conversions and profitability are stronger.
- Review telephone campaign costs before scaling that channel.
- Do not scale the test version yet; redesign or retest it.
- Use high-probability leads as the first follow-up audience.

What recruiters should notice

- Business problem defined before analysis.
- Multiple datasets transformed into a connected model.
- Statistical testing and ML are used with business interpretation.
- Dashboard is built for decisions, not only visuals.

Where AI assisted

- Business framing and KPI ideas.
- Python / SQL / DAX error review.
- Dashboard insight wording.

What I validated

- Jupyter notebooks executed manually.
- SQL Server tables and measures checked.
- Power BI visuals matched dashboard KPIs.

How to explain it

“AI was used as an analytics co-pilot, not as a replacement for analysis. I used it to improve framing, debug logic, and communicate insights clearly, while I executed and validated the final calculations myself.”

Current limitations

- A/B assignment is synthetic because the public dataset has no real campaign variant field.
- Cost and contribution values are assumptions for portfolio demonstration.
- Lead scoring uses a simple baseline logistic regression model.
- Dashboard is built locally, not connected to a production refresh pipeline.

Next version

- Replace assumptions with real CRM and finance data.
- Add automated refresh and data-quality checks.
- Compare multiple ML models and add lift/gain charts.
- Publish GitHub README and portfolio case study with screenshots.

30-second project pitch

“I built an end-to-end BFSI campaign analytics project using a real public banking dataset. I cleaned and transformed the data in Python, created a connected experiment and campaign-cost model, loaded clean tables into SQL Server, performed A/B testing and ML lead scoring, and built a two-page Power BI dashboard. The dashboard helps management understand conversion, cost efficiency, channel profitability, and which customers should be prioritized for future outreach.”

Built for junior data analyst portfolio: Python + SQL Server + Power BI + Business Communication